# IBM SkillsBuild Internship Project: Data Analytics & AI

## 📌 Project Overview

This project focuses on performing Comprehensive Data Analytics and Exploratory Data Analysis (EDA) on business datasets. It includes data preprocessing, statistical modeling, insights extraction, and data visualization to drive data-informed business decision-making. Developed as part of the **IBM SkillsBuild Internship Program** conducted by BharatCares in association with AICTE.

## 👤 Author Information

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- **Internship Track:** Data Analytics & AI
- **Organization:** BharatCares in association with AICTE
- **Platform:** IBM SkillsBuild

## 📄 Submitted File Details

As per internship guidelines, the following 4 files are generated for submission:

1. **Rohit_DataAnalyticsProject.py** - Complete Python source code for data cleaning, EDA, and visualization.
2. **requirements.txt** - List of Python dependencies required to run the project.
3. **Rohit_ProjectReport.md** - Full project documentation report.
4. **README.md** - Project overview, technology stack, and run instructions.

## 📊 Dataset Information

- **Dataset Name:** Business & Customer Performance Analytics Dataset
- **Source:** IBM SkillsBuild / Standard Analytics Dataset
- **Key Attributes:**
    - CustomerID: Unique identifier for each record
    - Age: Customer demographic metric
    - Annual_Income_k$: Income metrics in thousands of dollars
    - Spending_Score: Normalized score based on purchasing behavior
    - Category: Product categorization for segment analysis

## 🛠️ Technology Stack

- **Programming Language:** Python 3.x
- **Data Processing & Analytics:** pandas, numpy
- **Data Visualization:** matplotlib, seaborn

- **Machine Learning Tools:** scikit-learn
- **Environment:** Jupyter Notebook / Google Colab / VS Code

# 🚀 Setup & Execution Instructions

## 1. Environment Setup

Ensure Python 3.8+ is installed on your system.

## 2. Install Dependencies

Install all required libraries using the requirements.txt file:

```
pip install -r requirements.txt
```

## 3. Execution

Run the primary python script:

```
python Rohit_DataAnalyticsProject.py
```

*Submitted for IBM SkillsBuild Internship Project Evaluation by Rohit.*